Current Status and Future Prospects of Brain-Inspired Intelligence Development: A Review

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Abstract

Brain-inspired intelligence is a form of machine intelligence that mimics human brain activity. Achieving machine intelligence similar to human consciousness is the ultimate task of brain-inspired intelligence and the long-term goal of mankind. This paper provides an overview of the current status of the brain-inspired field from perspectives of core technologies, talent distribution, and industrial development. It highlights current issues in brain-inspired domain such as the shortage of professional technical talent, slow industrialization processes, inadequate fundamental theoretical research, and a lack of breakthroughs in key technologies. The paper suggests that we should focus on building a talent cultivation system for brain-inspired intelligence, integrating the concept of brain-inspired technologies into educational plans. Furthermore, it recommends advancing the integration of technology and application and strengthening the strategic planning of industries related to brain-inspired intelligence to play a significant role in the new round of development in human society.

CCS Concepts

• **Social and professional topics** → Professional topics; Computing education.

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Keywords

Brain-inspired intelligence, Brain-computer interface, The talent cultivation model, Planning and layout for industry

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1 INTRODUCTION

Brain-inspired intelligence represents a cutting-edge approach to artificial intelligence that seeks to replicate the complexities of human brain activity through a sophisticated interplay of software and hardware technologies [1]. This rapidly evolving field has gained prominence due to its potential to significantly enhance various domains. In neuroscience and cognitive science, these technologies offer deeper insights into brain function and cognitive processes, potentially leading to breakthroughs in understanding mental health conditions and neurological disorders. In the realm of artificial intelligence, brain-inspired systems could enhance the capabilities of machine learning algorithms, leading to more sophisticated and intuitive AI applications. Additionally, industries such as robotics, automation, and human-computer interaction stand to benefit from more adaptive and efficient intelligent systems. Therefore, research in brain-inspired intelligence is of paramount importance.

Currently, brain-inspired intelligence has achieved some technical breakthroughs. For instance, in hardware, there have been significant developments in designing neuromorphic chips that emulate neural networks, which enable more efficient data processing and energy savings compared to traditional computing architectures. On the software side, algorithms inspired by brain structures and functions, such as spiking neural networks, have shown promise in improving learning efficiency and adaptability.

However, the field still faces challenges, including overcoming technical obstacles, addressing shortages of skilled professionals, and integrating research advancements into practical applications. As a result, this article reviews the current state and future prospects of brain-inspired intelligence, focusing on core technologies, talent distribution, and industrial progress. It also discusses major challenges, such as the need for further technological breakthroughs and effective strategies for talent development. By providing strategic insights and recommendations, this article aims to advance the field and enhance its practical impact.

2 OVERVIEW OF BRAIN-INSPIRED INTELLIGENCE

Achieving artificial intelligence with human-like consciousness, inspired by the study of the human brain, is a long-term goal for humanity. Research in brain-inspired intelligence encompasses multiple levels of exploration [2, 3]. Firstly, there is the theoretical level, which includes research on brain mapping and cognition. For example, the University of Pennsylvania has launched a comprehensive brain development map covering the human lifespan, while China's Human Brain Project aims to establish an open resource for Chinese brain imaging. These initiatives provide essential data foundations for researchers. Additionally, cognitive studies are crucial, as measuring and investigating activities in areas such as the thalamus and primary somatosensory cortex help in understanding brain function mechanisms.

On the hardware front, research institutions and companies worldwide have developed various brain-inspired chips. Notable examples include IBM's TrueNorth [4], Intel's Loihi [5], and Tsinghua University's Tianjic. These chips are designed to emulate neuron functionalities, thereby providing hardware support for brain-inspired intelligence.

In terms of software, numerous frameworks and platforms simulate brain functions. Brain Py 2.0 integrates model definition, simulation, and training in a streamlined and efficient manner. Other frameworks like BindsNet, the spiking neural network-based simulation computing platform Nengo, and the fully spiking neural network-based brain-inspired cognitive engine "Zhimai" [6] also exhibit capabilities in brain-inspired AI models and the simulation of brain functions and structures, thus offering technical support for brain-inspired intelligence research and applications.

On the application level, major tech companies and research institutions have launched brain-inspired intelligence projects. Examples include Google Brain in 2012, iFlytek's Hyper Brain in 2014, JD.com's JD Brain in 2015, and Baidu Brain, which was showcased at the 2016 Baidu World Conference. These projects aim to apply brain-inspired intelligence technologies to fields such as speech recognition, image processing, and intelligent recommendation systems, thereby enhancing AI performance and efficiency.

In summary, the brain-inspired intelligence field encompasses multiple aspects including theory, hardware, software, and applications. Its development requires interdisciplinary cooperation and multifaceted efforts. Through a deep understanding and simulation of brain mechanisms, brain-inspired intelligence has the potential to surpass the limitations of traditional computational intelligence and achieve more advanced AI systems.

3 CURRENT STATE OF DEVELOPMENT IN BRAIN-INSPIRED DOMAIN

3.1 Core technologies

Brain-inspired intelligence aims to achieve three levels of goals: mimicking the brain's structural level, approximating the device level and surpassing the functional level. Its technical system is divided into four layers: foundational theory, hardware, software, and products [7]. The foundational theory layer focuses on brain cognition and neural computing, exploring mechanisms like brain plasticity and functional structure. The hardware layer develops neuromorphic chips using non-von Neumann architectures, including spiking neural network chips and memristors. The software layer encompasses core algorithms and general technologies, such as weakly supervised learning, spiking neural networks [8], reinforcement learning, and adversarial neural networks. The product layer includes interactive products like brain-machine interfaces and brain-inspired robots [9].

China has made significant progress in core technologies within the brain-inspired domain. In foundational theory, numerous institutions and universities conduct research in brain science, cognitive science, and neural networks. For example, Professor Shen Dinggang of ShanghaiTech University has pioneered medical imaging AI research, applying deep learning to brain development and disease analysis. Professor Han Junwei's BRAIN Lab at Northwestern Polytechnical University focuses on brain cognition and bio-inspired intelligent computing methods [10]. In hardware, significant breakthroughs have been made in brain-inspired computing chips. The Brain-Inspired Computing Research Center at Tsinghua University developed "Tianjic," a heterogeneous fusion brain-inspired computing chip integrating artificial and spiking neural networks. They proposed "brain-inspired computing completeness" and a decoupled software-hardware architecture, demonstrating hardware completeness and expanding brain-inspired computing applications. In software, the Institute of Automation at the Chinese Academy of Sciences developed the BrainCog brain-inspired cognitive intelligence engine based on spiking neural networks. This open-source platform integrates biological neurons, encoding strategies, and learning rules, supporting brain-inspired AI and multi-scale brain simulations. In summary, China has established a strong and competitive position in the global brain-inspired technology landscape, with several research teams achieving international recognition for their pioneering work.

3.2 Talent distribution

From the perspective of the geographical distribution of talents, we can gain insight from the entities applying for patents in the brain-inspired intelligence domain. As shown in Figure 1, the top 10 global entities applying for patents related to brain-inspired intelligence are primarily located in the United States (7 entities), South Korea (1 entity), and China (2 entities) [11]. Of these, 8 are corporate entities, while the remaining 2 are academic institutions. This distribution indicates that, on a global scale, talent in the brain-inspired intelligence field is primarily concentrated in countries with strong technological and research capabilities, particularly in the United States, which dominates the landscape

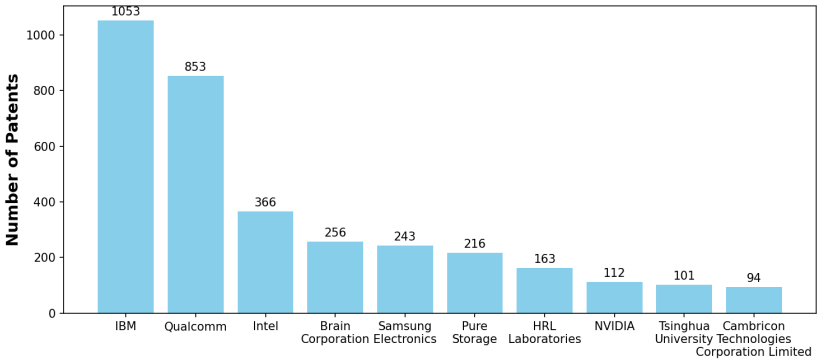

Figure 1: Top 10 entities for brain-inspired intelligence technology patent applications around the world (number of applications) as of October 2021.

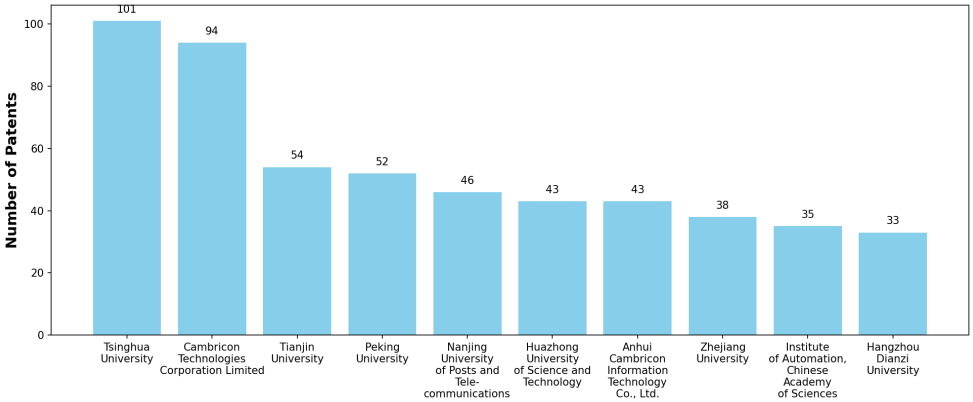

Figure 2: Top 10 entities for brain-inspired intelligence technology patent applications in China (number of applications) as of October 2021.

with a significant number of leading companies and institutions. Furthermore, the dominance of corporate entities in patent applications suggests that much of the talent is oriented towards industrial and practical applications of brain-inspired technologies, focusing on commercial innovation and product development. Meanwhile, academic institutions, though fewer in number, play a critical role in foundational research and the training of new experts in the field. This distribution reflects the collaborative nature of the field, where academia and industry work together to advance brain-inspired intelligence.

Talent in China's brain-inspired domain is primarily concentrated in major cities and technological innovation centers. Figure 2 shows the top 10 entities in China applying for brain-inspired intelligence technology patents as of October 2021 [11], predominantly universities and research institutions in Beijing, Shanghai, Shenzhen, and Hangzhou. This indicates a heavy concentration of talent in the eastern regions, reflecting an uneven distribution across the country, with a notable focus on developed urban areas and innovation hubs.

From the perspective of academic backgrounds, talent in the brain-inspired domain spans multiple disciplines, including computer science, neuroscience, psychology, mathematics, and physics. Researchers from different fields contribute their expertise: computer science and engineering professionals develop brain-inspired computational models, algorithms, and software systems; experts in neuroscience and biology study the brain's structure and functions, providing a biological foundation for brain-inspired technology; and researchers in mathematics and physics participate in the establishment and analysis of brain-inspired models, offering theoretical support for brain-inspired intelligence research.

3.3 Industrial development

The development of brain-inspired intelligence technology is still largely in the research phase, while Brain-Computer Interface (BCI) technology has seen partial commercialization. BCIs connect the brain to external devices, using brain signals to control systems. As shown in Figure 3, according to a report by the international market research agency Verified Market Research, the global brain-inspired intelligence market was valued at $1.514 billion in 2020 and is projected to reach $14.048 billion by 2030, with a compound annual

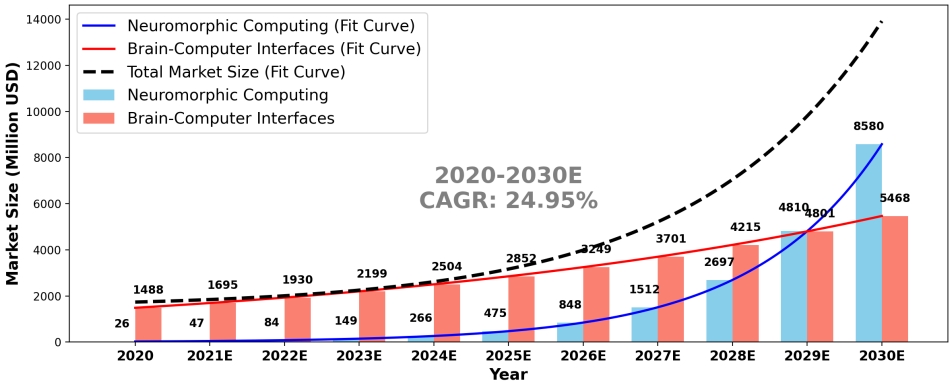

Figure 3: The global brain-inspired intelligence market size and forecast from 2020 to 2030.

growth rate (CAGR) of 24.95%. Currently, BCI holds the largest market share within this sector [12]. To analyze the future trends of the brain-inspired intelligence market, we applied an exponential growth model to fit the development curves of Neuromorphic Computing and BCI. The exponential growth model is described by the following equation:

$$y = a \cdot e^{b(x-x_0)} + c \tag{1}$$

Where x_0 represents the reference or baseline year from which the growth is calculated, a represents the initial value, b is the growth rate, and c is an adjustment parameter for the curve.

The fitted data indicates that both Neuromorphic Computing and BCI markets exhibit exponential growth. Neuromorphic Computing is projected to grow from $26 million in 2020 to $8.58 billion by 2030, representing a significant increase. In comparison, the BCI market is expected to grow from $1.488 billion in 2020 to $5.468 billion in 2030, showing steady but slower growth. The rapid growth of Neuromorphic Computing can be attributed to advances in technology and increasing market demand, particularly in areas such as chip design and machine learning, where the technology has broad applications. In contrast, BCI's development has been slower due to several factors, including the complexity of the technology, legal and ethical constraints, and the more limited scope of its current applications. BCI primarily serves specific areas like medical rehabilitation, where it requires highly precise sensors to capture brain signals and convert them into actionable commands. This makes commercialization slower due to the technical challenges and the stringent regulatory and ethical scrutiny that BCI faces. Despite the slower growth, BCI still demonstrates steady potential, and the overall brain-inspired intelligence market shows strong expansion, signaling the vast potential of the brain-inspired intelligence sector over the next decade.

The large-scale application of brain-inspired technology will significantly change human society and industrial patterns, as evidenced by the commercialization of BCI technology mentioned above. These applications can not only increase productivity and improve quality of life but also bring new business models and industrial opportunities. China is further strengthening the collaboration between industry, academia, and research institutions to promote the commercialization and industrialization of brain-inspired technology. Figure 4, released by Verified Market Research, shows the global distribution of the brain-inspired intelligence market in 2020. Overall, the United States holds a dominant position in the industry, with China closely following, achieving remarkable progress [13].

4 CHALLENGES IN THE DEVELOPMENT OF BRAIN-INSPIRED INTELLIGENCE SECTOR

4.1 Significant talent gap

Currently, the talent cultivation model and system in the brain-inspired domain are quite ambiguous, resulting in a substantial talent gap [14]. Universities, the primary sources of talent, face significant challenges in both undergraduate and postgraduate education. The design of curriculum systems and training programs for brain-inspired fields is crucial, but most top-tier universities have yet to establish dedicated brain-inspired programs for targeted training at the undergraduate level. Brain-inspired content is primarily covered under artificial intelligence (AI) programs, but these AI programs touch on brain-inspired knowledge only superficially.

From a recruitment standpoint, hiring new employees and talent from interdisciplinary fields is challenging. Despite the recognized demand for interdisciplinary expertise in HR departments, finding suitable candidates remains difficult. Often, the professional focus of applicants differs from the keywords provided in job descriptions. Given the broad scope of the brain-inspired research field, identifying the right keywords and professional focus is particularly challenging. As this discipline is relatively new, there is an urgent need to cultivate and produce new talent.

4.2 Slow pace of industrialization

Although researchers are confident about the development prospects of brain-inspired technology, its industrial application still faces numerous challenges. Significant breakthroughs in the application of brain-inspired technology have yet to be achieved, and its industrialization requires more time and in-depth development. Within the entire brain-inspired industry, the hardware and

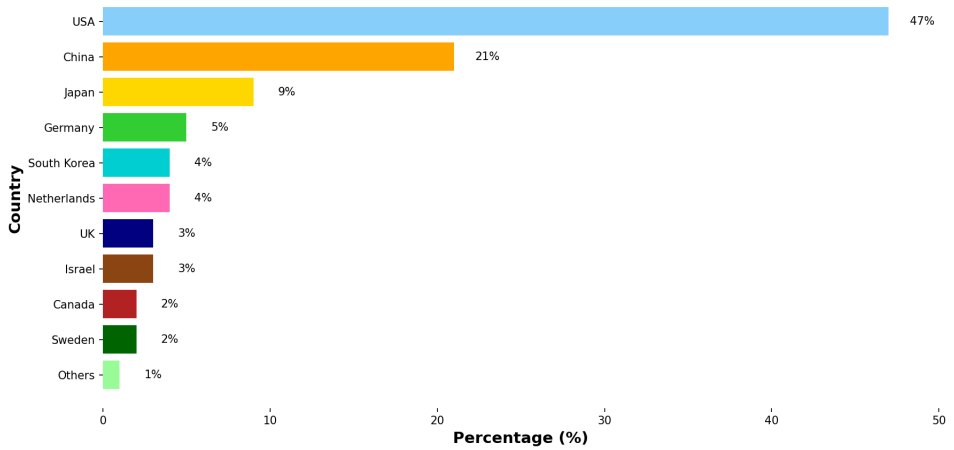

Figure 4: Global brain-inspired intelligence market distribution in 2020 (by Market Share).

chip sectors are more active. For instance, there have been numerous breakthroughs in the technical architecture and application of in-memory computing. Investors are more inclined to invest in the hardware sector due to its higher barriers to entry compared to algorithms. Consequently, most startups in the brain-inspired industry focus on chip development. The field of brain-inspired research has not yet produced a "killer application," which is crucial for industrial development. Previous studies have questioned the necessity of using spiking neural networks or brain-inspired neurons over existing technologies like deep learning. Therefore, the brain-inspired industry still requires long-term development and nurturing to realize its full potential.

4.3 Insufficient basic research and key technological breakthroughs

In brain-inspired field, critical bottlenecks include insufficient basic theoretical research and a lack of breakthroughs in key technologies. Despite progress in hardware and software, our understanding of the brain's structure and functional mechanisms is still limited, hindering the development of technologies that can truly emulate brain functions.

Key areas like brain-inspired computing chips and cognitive engine software platforms have not yet seen significant breakthroughs. Initial achievements in approximating brain functions at the hardware level and surpassing brain capabilities at the software level are notable, but there's still a considerable gap compared to a full brain simulation. True brain-inspired computing chips have yet to be realized, and further research and innovation are needed. Moreover, cognitive engine software platforms and other crucial technologies require more investment and breakthroughs to achieve a true simulation and realization of brain-inspired intelligence.

5 FUTURE DIRECTIONS AND STRATEGIC RECOMMENDATIONS FOR BRAIN-INSPIRED TECHNOLOGY DEVELOPMENT

5.1 Promoting brain-inspired technology: coordination, curriculum development, and interdisciplinary talent

5.1.1 Coordination between research and education. In the brain-inspired field, coordination between research and education is essential [15]. First, research in brain-inspired technology aims to break existing computational paradigms, requiring highly interdisciplinary collaboration. In education, integrating brain-inspired technology concepts into educational programs is crucial. For example, Peking University has established an Intelligent Science and Technology program that combines computer science with cognitive science, aiming to incorporate biological cognitive systems into computer science. This program allows undergraduate students to understand brain operations and gradually delve deeper into this field. Both in research and education, sustained scientific exploration and the continuous emergence of new disciplines and training systems are crucial for the advancement of brain-inspired technology.

5.1.2 Importance of undergraduate and graduate curriculum development. In education, designing brain-inspired curriculum systems and training programs is essential. Most top-tier universities in China have not established dedicated brain-inspired programs at the undergraduate level. Brain-inspired content is primarily included within AI programs, but often with limited coverage. Similarly, many international universities also lack standalone brain-inspired programs, instead integrating relevant content into broader AI or neuroscience courses. Developing these programs requires foundational courses in mathematical fundamentals of AI, courses on brain cognition, neuroscience, and bioinformatics, and core professional courses on specific brain-inspired topics. Application technology

courses and practical courses should also be included to stimulate student interest.

5.1.3 The need for interdisciplinary talent cultivation. In interdisciplinary education, balancing three aspects is crucial. First, equal emphasis should be placed on theoretical knowledge and programming practice. Students from fields like physics, chemistry, and mathematics need to develop programming skills to build practical concepts and computational thinking. Second, the curriculum should balance neuroscience and computational science to ensure sufficient interdisciplinary integration. Lastly, a balance between theoretical learning and cutting-edge technology is essential. For instance, Peking University offers an undergraduate course titled "Computational Foundations of Neural Networks," designed for students from various disciplines, including computer science, psychology, physics, mathematics, and even chemistry and materials science. This course aims to provide foundational knowledge and skills necessary for advancements in brain-inspired technologies.

5.2 Industrial advancement and application scenarios for brain-inspired technology

5.2.1 Promising Development Prospects and Strategic Timing. From a consumer perspective, the development prospects of brain-inspired technology products are very promising [16]. For instance, spiking chips, known for their low power consumption, can be integrated into electronic devices such as smartphones. The development of low-power chips and their underlying algorithms will be crucial not only for consumer applications like handheld devices but also within the broader IoT framework. As research progresses, brain-inspired technology holds vast potential in the consumer sector. Therefore, although brain-inspired research is currently in its early stages, the advent of related applications will create significant competitive advantages. Missing the optimal timing could result in a loss of competitive edge, as evidenced by the emergence of ChatGPT, which allowed OpenAI to seize a large portion of market opportunities. Therefore, advance planning and strategic layout are crucial to maintaining a competitive advantage.

5.2.2 Combining Relatively Mature Technologies with Corresponding Application Scenarios. As a cutting-edge technology, brain-inspired technology has considerable value and potential future applications. However, it may still be in the exploratory stage within laboratories or have ambiguous application scopes. Combining relatively mature technologies with corresponding application scenarios is a promising direction. For example, integrating brain-machine interface technology with traditional rehabilitation methods could create innovative products that address existing deficiencies, thereby driving industry progress. Expansion of brain-inspired technology can also focus on similar technologies within the artificial intelligence field, such as ChatGPT, which has sparked widespread interest and holds extensive potential value in applications like companionship and communication for the elderly.

6 CONCLUSION

This paper provides a comprehensive overview of brain-inspired intelligent technology, covering concepts, theories, hardware, software, and applications. It reviews the current state of the brain-inspired field, examining core technologies, talent distribution, and industrial development. The analysis reveals significant challenges, including a serious talent shortage due to an ambiguous training system and a slow industrialization process caused by limited breakthroughs in core technologies.

To address these issues, the paper emphasizes the need for better coordination between scientific research and education to enhance the talent training system. Additionally, it advocates for integrating mature technologies with application scenarios to promote industrial planning and advancement. By tackling these challenges, the aim is to foster a more innovative and prosperous brain-inspired intelligence industry worldwide.

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